SAMPLE

Cosmic Explorers: Commercial Space

The New Space Race

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Free sample: Introduction & Chapter 1

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Introduction

For the first four decades of the space age, going to orbit meant going through a government. Rockets were national assets, flown by national agencies, carrying payloads that served national purposes. If a company wanted a satellite in orbit, it bought a ride on a state's rocket and waited its turn. The idea of a market in space, of firms competing to sell launches the way airlines sell seats, sounded like science fiction.

It no longer does. Today a private company flies most of the mass humanity sends to orbit, landing its boosters on barges and reflying them dozens of times each. Private operators run satellite fleets numbering in the thousands, sell imagery of any patch of Earth, carry paying passengers past the edge of space, and are building their own stations to replace the government one. The money, the engineering, and increasingly the decisions about what happens in space have shifted decisively toward the commercial world.

This book tells that story: how commerce got into orbit, what the space industry actually sells, who the major players are and how they got there, and what a functioning space economy looks like when you strip away the hype. A definition is worth fixing at the outset, because the words drift. In this book, commercial space means activity in space, or in direct service of it, that is undertaken for profit by private actors: companies that build, launch, insure, operate, and use spacecraft, and the markets that form around them. It does not mean every dollar spent on space, since governments remain the industry's largest customers by far; it means the portion of spaceflight that answers to invoices rather than appropriations.

The border between the two is the story's most interesting terrain, and the book keeps returning to it: nearly every great commercial success in orbit began life as a government requirement, and nearly every great commercial failure began as a business plan that assumed the government's market was the market.

It is a business history, a technology survey, and an honest look at the risks, all in one volume.

The arc falls into three eras. First came the satellite era, in which the only profitable things in space were communications relays and the rockets that served them. Then came the disruption era, when new companies with new philosophies, above all reusability, broke the launch monopoly and collapsed the price of reaching orbit. Now comes the build-out era, in which the industry is trying to do more than cheapen the trip: it wants to build industry, tourism, and settlement in space itself.

Along the way we will meet the Cold War telephone companies that quietly founded the space business, the European consortium that turned launch into a product, the internet millionaire who decided rocketry's problem was accountancy, the constellation builders whose satellites now outnumber all other spacecraft combined, and the investors who have poured hundreds of billions into a sector whose revenues, so far, remain a fraction of that sum.

We will also be honest about the difficulties. The commercial space story includes spectacular bankruptcies, missions that ended in craters, markets that never materialized, and an orbital environment that the industry's own success is now threatening. A clear-eyed account makes the achievements more impressive, not less.

What you hold is a map of the greatest industrial frontier opened in living memory. The frontier is real, the maps are still being drawn, and the survey has only begun.

The subject goes first, before the chapters begin, because the commercial space economy's headline growth conceals its structure. The industry's revenue is dominated, in the ledger the final chapters will audit, not by launch or tourism but by satellites serving customers on the ground, communications, navigation, and observation, and the celebrated rockets and stations that fill the news are the visible tip of an industry whose actual product is information about Earth. The reader should hold that inversion from the first page, because every chapter that follows is the story of how a government technology became an economy, and the honest version of that story runs through balance sheets, not just launches.

A second framing observation belongs beside it, the one that separates this book from the boosterism it could easily have become. The commercial space age was not announced; it accumulated, in a series of bankruptcies, rescues, contracts, and price collapses spread over two decades, and the industry's current shape is the survivor of more failed companies than successful ones. The reader should treat that history as the book's spine, because the future chapters' forecasts can only be judged against the pattern the early chapters establish, that in the space business the graveyard is as instructive as the launchpad.

Chapter 01: From Government Program to Industry

Commerce entered space through the telephone. The first satellites were scientific and military, but the first satellite business was communications, and its logic was inescapable: a relay station 36,000 kilometers up could see half the planet at once, replacing a thousand undersea cables and microwave hops for transoceanic traffic. Governments wanted the capability, but telephone traffic was a commercial commodity, and so the earliest space companies were born as partners of the state rather than its contractors alone.

The American model began in 1962, when Congress created Comsat, a shareholder-owned corporation chartered to represent the United States in a global satellite consortium. That consortium became Intelsat, an international cooperative whose early bird spacecraft proved in 1965 that transoceanic satellite telephony was not merely possible but profitable. Within a decade, calls routed through space had become ordinary, and the revenue from those calls was building an industry. The Soviet bloc built its own system, and Canada, with its vast empty distances, built the first domestic satellite network, showing that a single nation could justify a space system on purely commercial grounds.

The lesson of that first generation was that space businesses grow best in the sheltered ground of monopoly and treaty. Intelsat's signatories were national post and telephone monopolies, guaranteeing both traffic and capital. The spacecraft were built by aerospace giants, insured by a specialized market, and launched by governments. Risk was pooled, prices were regulated, and the whole arrangement worked, after its fashion, for three decades.

Sheltered ground eventually became a cage. When telecommunications was deregulated in the 1990s, the old consortium was privatized into a competitive firm, and suddenly satellite operators had to fight for traffic against fiber optic cables, which were burying the oceans in cheap bandwidth. The first space boomlet, the great low-earth-orbit telephone constellations of the 1990s, met precisely this fate: their business plans were drawn in a world without cheap fiber, and by the time the satellites were built, the world had changed underneath them.

Meanwhile the launch side of the young industry was following its own path. Through the 1970s and 1980s, flying to orbit meant buying a ride on an American or Soviet rocket, at prices and on schedules set by governments. Europe, having no civilian launcher of its own and watching the United States fly commercial payloads on the Shuttle, decided in the 1970s to build one, and

in 1980 did something with no precedent: it formed Arianespace, a company whose product was the launch itself, sold to any customer who could pay.

The company's timing turned out to be extraordinary. The Shuttle was grounded by the 1986 accident, American expendable rockets were restricted and expensive, and suddenly the only reliable Western commercial launcher belonged to a firm that treated spaceflight as a product line rather than a program. The Shuttle interlude deserves its footnote, because it taught the industry the danger of a monopoly with good intentions. The American plan of the early 1980s was to fly commercial and foreign payloads on the Shuttle alongside government ones, putting the entire Western launch market on a single government system, and the market obeyed until the 1986 accident grounded the fleet and left customers stranded with satellites and no ride. The expendable vehicles scrambled back into production under license, the Europeans banked a decade of orders, and the lesson was written into every subsequent national space policy: no launch monopoly, however splendid, is ever to be trusted with the whole market again. Commercial space owes part of its existence to that single, costly correction.

Ariane rockets captured half the open market, and the lesson sank in across the industry: the vehicle was not the product. The delivery was.

By the early 2000s the pieces of a genuine sector existed: commercial satellite operators, a commercial launch company, a specialized insurance market in London, and a small crowd of manufacturers selling spacecraft to anyone. Global space revenues passed the tens of billions, with communications the overwhelming majority. But the sector remained, in character, a government guest: its biggest customer was still the state, its prices assumed the state's cost structures, and its rockets were still thrown away after a single use, exactly as they had been since the 1950s.

One more strand of the old order deserves remembering: the goods themselves were becoming merchandise even while the rides stayed governmental. By the 1980s, satellite manufacturers in the United States and Europe were selling spacecraft outright to private operators and foreign governments alike, and a secondary industry of ground stations, transponder brokers, and launch insurance grew up around the fleet. The satellite had become a machine you could simply buy, in a catalog of buses and payloads priced like ships, and that commercial normalcy around the payload made the government monopoly on the ride look stranger every year. The market was complete at both ends of the transaction; only the middle, the delivery, remained a state affair, and monopolies in the middle of markets do not last.

That wastefulness was the crack through which the future entered. A rocket

that burns a hundred million dollars of hardware on every flight can never be more than a luxury good. The people who saw this clearly were dismissed for decades, and the commercial space age in the modern sense began the day someone with the money and patience to test the idea put real hardware on the answer. That story, the reusable revolution, is the next chapter's subject.

The second product's false start belongs in this chapter too, because it shows the model's limits a decade before anyone named them. Earth imaging was born government, first as reconnaissance and then, from 1972, as the civilian Landsat program, whose openly available imagery of crops, forests, and geology was quietly seeding a future industry of analysts. In the mid-1980s the United States tried to hand the system to a private operator, on the theory that a commercial data product should have a commercial owner, and the result was instructive: the contractor inherited a satellite, a data archive, and a customer base consisting mostly of other government agencies, priced the data high enough to recover its costs, and watched the user community shrink as researchers and foreign agencies turned to free alternatives.

Within a decade the system was back in government hands, by which time the lesson was legible to anyone who read it: a market cannot be privatized before it exists, and the products that thrive commercially in space are the ones whose customers are many and private, not the ones whose single customer is the state. The imaging industry that eventually did emerge, the subject of a later chapter, was built the other way around, from small cheap spacecraft outward, and its founders cited the Landsat affair as their founding caution.

End of free sample. The full book is available at putrabooks.github.io